

PG-DSL: T1 Vectorization for Composition-Aware Admission

Theorem Framework (Mission 3A addendum)

Abstract

This document extends the Mission 1A theorem framework (`mission_1a/theorems.tex`) with a vectorized form of T1, denoted $T1_v$, accommodating the canonical PG-DSL admission layer's expanded admission domain introduced by Mission 3B (Z3b option B, `dec_01KR49FTB2ND38DW36094G95ZM`). $T1_v$ replaces the scalar physical-state distance δ with a per-coordinate vector δ_v on a state space $\Sigma \subseteq \mathbb{R}^n$ and a max-norm threshold $\|\delta_v\|_\infty$. The Mission 1A T1 is recovered as the $n = 1$ specialization. T2 (lifting soundness on a structured class) is unchanged. T3 (strict composition with runtime CPS-IDS) retains its strict-superset claim; only the witness partition shifts ($Z_3 \rightarrow A_{\text{static}}, W_1 + W_2 \rightarrow A_{\text{runtime}}$ per `dec_01KR49FTB2ND38DW36094G95ZM`). A two-tank tractable model (`mission_3a/tractable_model.v.md`) and verification script (`mission_3a/t1_v_verify.py`) corroborate the bound numerically; the Z3 composition witness from Mission 3B sits provably in the detected region of $T1_v$.

1 Notation and Vectorization

Let $\Sigma \subseteq \mathbb{R}^n$ denote the physical state space relevant to admission verification, with state coordinates indexed by $k \in \{1, \dots, n\}$. For the canonical SWaT P1+P2 substrate, state coordinates of interest at admission time are the level coordinates (L_{T101}, L_{T201}) ($n = 2$); the scalar Mission 1A model is the $n = 1$ projection.

Admission target. Under canonical PG-DSL (Mission 3B Option B, `dec_01KR49FTB2ND38DW36094G95ZM`), the admission target T ranges over either a single tool description D_i or an ordered pair of tool descriptions (D_i, D_j) . The lifter L_{verify} is unchanged (per-tool); the composition layer replays ordered pairs in the digital twin (DT) over a fixed horizon and an enumerated set of canonical initial states.

Lifted vectorized claim. The lifter produces, for each coordinate k , an admissible region $A_\varphi(k) \subseteq \mathbb{R}$ (or $\top$ if φ_v does not name coordinate k). Composition admission additionally supplies an effective operating safe band $A_{\text{eff}}(k)$ from runtime safety knowledge; in practice $A_{\text{eff}}(k) = A_\varphi(k)$ when φ_v names a band, else $A_{\text{eff}}(k)$ is the externally-provided safe band for coordinate k . For clarity, $T1_v$ is stated against $A_{\text{eff}}(k)$.

Per-coordinate discrepancy. Define the per-coordinate trajectory discrepancy of the actual physical trajectory $\psi_v^* : [0, T_{\text{horizon}}] \rightarrow \Sigma$ from the admissible region as

$$\delta_v[k] := \sup_{t \in [0, T_{\text{horizon}}]} d_k(\psi_v^*(t)[k], A_{\text{eff}}(k)), \quad d_k(x, [a, b]) := \max(0, x - b, a - x),$$

with $d_k(x, \top) := 0$. Then $\delta_v = (\delta_v[1], \dots, \delta_v[n]) \in \mathbb{R}_{\geq 0}^n$. The vectorized magnitude reported to the matcher is the max-norm $\|\delta_v\|_\infty = \max_k \delta_v[k]$.

Per-coordinate error bounds. Let $\varepsilon_{L,v} \in \mathbb{R}_{\geq 0}^n$ denote the per-coordinate worst-case lifter semantic error on a structured class $\mathcal{C}$ (analogue of the scalar ε_L from Mission 1A, indexed per coordinate). Let $\varepsilon_{DT,v} \in \mathbb{R}_{\geq 0}^n$ denote the per-coordinate per-sample DT fidelity error. The per-coordinate matcher tolerance is $\varepsilon_v[k] := \varepsilon_{L,v}[k] + \varepsilon_{DT,v}[k]$.

2 T1_v — Vectorized Detection Threshold

Theorem 1 (T1_v, Vectorized Detection Threshold). *Let L_{verify} have vectorized semantic error $\varepsilon_{L,v}$ on class $\mathcal{C}$ and let DT have vectorized per-sample fidelity error $\varepsilon_{DT,v}$ (no Gaussian tail per coordinate). Set the vectorized matcher threshold $\tau_v^* := \|\varepsilon_{L,v} + \varepsilon_{DT,v}\|_\infty$. For any admission target T (single tool or ordered tool pair) with lifted vectorized claim φ_v and actual trajectory-projected physical effect ψ_v^* whose vectorized discrepancy δ_v satisfies $\|\delta_v\|_\infty \geq \tau$ for some $\tau > 0$, PG-DSL’s admission-time detection probability satisfies*

$$\Pr(\text{detect} \mid \|\delta_v\|_\infty \geq \tau) \geq 1 - F_v(\tau, \varepsilon_{L,v}, \varepsilon_{DT,v}),$$

where $F_v : \mathbb{R}_{\geq 0} \times \mathbb{R}_{\geq 0}^n \times \mathbb{R}_{\geq 0}^n \rightarrow [0, 1]$ is monotone non-increasing in τ . Under bounded-error sensing,

$$F_v(\tau, \varepsilon_{L,v}, \varepsilon_{DT,v}) = \begin{cases} 1 & \text{if } \tau < \delta_v^* \\ 0 & \text{if } \tau \geq \delta_v^* \end{cases}, \quad \delta_v^* = c_t \cdot \tau_v^* = 2 \|\varepsilon_{L,v} + \varepsilon_{DT,v}\|_\infty,$$

with $c_t = 2$. Below τ_v^* the matcher cannot fire deterministically; below $\delta_v^* = 2\tau_v^*$ adversarial bounded-error noise can drive the empirical max-norm under the threshold and detection is not guaranteed.

Proof sketch. Per-coordinate triangle inequality, combined via max-norm.

Per-coordinate geometry. Fix the description (or pair) T . For each coordinate k where φ_v supplies an admissible region $A_\varphi(k)$ and the lifter’s output yields predicted region $A_{\text{lifted}}(k)$ within $\varepsilon_{L,v}[k]$ of the ground-truth lifted intent (Hausdorff sense), the lifter’s per-coordinate semantic error is bounded by $\varepsilon_{L,v}[k]$. The DT-measured trajectory $\psi_v(t)[k]$ lies within $\varepsilon_{DT,v}[k]$ of the actual plant trajectory $\psi_v^*(t)[k]$ for all t .

Triangle inequality on coordinate k . The DT-measured per-coordinate discrepancy $\hat{\delta}_v[k] := \sup_t d_k(\psi_v(t)[k], A_{\text{lifted}}(k))$ satisfies

$$\delta_v[k] - \varepsilon_{L,v}[k] - \varepsilon_{DT,v}[k] \leq \hat{\delta}_v[k] \leq \delta_v[k] + \varepsilon_{L,v}[k] + \varepsilon_{DT,v}[k].$$

Matcher firing rule. The matcher fires iff there exists a coordinate k with $\hat{\delta}_v[k] > \varepsilon_v[k] = \varepsilon_{L,v}[k] + \varepsilon_{DT,v}[k]$, equivalently $\|\hat{\delta}_v\|_\infty > \tau_v^*$.

Detection guarantee. If $\|\delta_v\|_\infty \geq \tau$ and the achieving coordinate k^* has $\delta_v[k^*] = \|\delta_v\|_\infty \geq \tau$, the lower bracket gives $\hat{\delta}_v[k^*] \geq \tau - \varepsilon_v[k^*]$. Detection is guaranteed when $\tau - \varepsilon_v[k^*] > \varepsilon_v[k^*]$, i.e., $\tau > 2\varepsilon_v[k^*]$. Taking the worst-case coordinate gives $\tau > 2\|\varepsilon_v\|_\infty = \delta_v^*$.

Impossibility region. If $\|\delta_v\|_\infty < \tau_v^*$, then for every coordinate k , $\delta_v[k] < \varepsilon_v[k]$. The upper bracket gives $\hat{\delta}_v[k] \leq \delta_v[k] + \varepsilon_v[k] < 2\varepsilon_v[k]$, but the matcher only requires $\hat{\delta}_v[k] > \varepsilon_v[k]$ to fire on coordinate k , so firing is possible on adversarial-friendly noise but not guaranteed for any. Between τ_v^* and $\delta_v^* = 2\tau_v^*$ the detection rate is empirically smooth (matcher fires for benign-side noise realizations and fails for adversarial-side ones).

The bounded-error step form of F_v follows: $F_v \equiv 1$ for $\tau < \delta_v^*$ and $F_v \equiv 0$ for $\tau \geq \delta_v^*$. □

Specialization to T1 (consistency check). At $n = 1$, $\|\delta_v\|_\infty = \delta_v[1] = \delta$, $\tau_v^* = \varepsilon_{L,v}[1] + \varepsilon_{DT,v}[1] = \varepsilon_L + \varepsilon_{DT}$, and $\delta_v^* = 2(\varepsilon_L + \varepsilon_{DT})$. F_v reduces to Mission 1A’s scalar F . $T1_v$ strictly generalizes T1.

Concrete instantiation (PI-authorized, brain to validate). On the two-tank tractable model (`mission_3a/tractable_model_v.md`), we anchor $\varepsilon_{L,v} = (0.116, 0.116)$ %-full per coordinate (Req2LTL accuracy gap, both coordinates) and $\varepsilon_{DT,v} = (0.10, 0.10)$ %-full (Mission 3B per-sample sensor floor). Then

$$\tau_v^* = \|\varepsilon_{L,v} + \varepsilon_{DT,v}\|_\infty = 0.216 \text{ %-full}, \quad \delta_v^* = 2\tau_v^* = 0.432 \text{ %-full}.$$

3 Witness Construction — Z3 in the Detected Region

The Z3 witness from Mission 3B (`mission_2a/witnesses/run_witness_z3.py`, `mission_3b/attacks/z3_simulator`) is a composition-only attack on the two-tank substrate.

Setup. Initial state $(L_{T101}, L_{T201}) = (90, \text{nominal})$, $V_{\text{couple}} = \text{closed}$, all P1 actuators idle. Admission target is the ordered pair (`open_valve_MV101`, `start_pump_P101`). Each constituent admits per-tool: lifted φ_v contains only single-actuator clauses $\text{actuator}(V_{\text{in1}}) := \text{open}$ and $\text{actuator}(P_1) := \text{on}$; per-tool DT replay from canonical MID initial state shows zero level effect (the prerequisite actuator is unset).

Composition replay. The Mission 3B canonical layer (`mission_3b/admission_layer/composition_verifier`) replays the ordered pair from each of $\{\text{LOW}, \text{MID}, \text{HIGH}\}$. Under HIGH initial state $L_{T101} = 90$, with $V_{\text{couple}} = \text{closed}$, net inflow drives L_{T101} at rate 1.0 %-full/s; the trajectory exits the safe band $A_{\text{eff}}(L_{T101}) = [25, 95]$ at $t = 11$ s and reaches the OVERFLOW predicate $L_{T101} \geq 99.5$ at $t = 19$ s.

Vectorized discrepancy.

$$\delta_v[L_{T101}] = \max_t d(L_{T101}(t), [25, 95]) = 99.5 - 95 = 4.5 \text{ %-full}, \quad \delta_v[L_{T201}] = 0 \Rightarrow \|\delta_v\|_\infty = 4.5 \text{ %-full}.$$

Detection verdict under $T1_v$. $\|\delta_v\|_\infty = 4.5 \gg \delta_v^* = 0.432 \text{ %-full}$ ($\sim 10.4\times$ above threshold). $T1_v$ predicts $F_v(4.5, \varepsilon_{L,v}, \varepsilon_{DT,v}) = 0$, i.e., $\Pr(\text{detect}) \geq 1$ under any bounded-error noise realization. Mission 3B empirical: Z3 caught at admission via composition pair on HIGH initial state, consistent with $T1_v$. $\square$

4 T2 — Unchanged

The T2 statement (lifting soundness on class $\mathcal{C}$, `mission_1a/theorems.tex` §T2) is unaffected by the admission-domain expansion. T2 governs how single tool descriptions lift to formal claims; composition admission operates on already-lifted single-tool claims and does not change the lifting step. The Mission 1A bound $\delta_{\text{lift}} \leq \delta_{\text{cov}}(\mathcal{C}, L_{\text{verify}}) + \delta_{\text{gram}}(\mathcal{C})$ remains the per-tool soundness statement.

5 T3 — Strict Composition with Annotated Witness Partition

The T3 statement (`mission_1a/theorems.tex` §T3) is unchanged: PG-DSL composed with a runtime CPS-IDS detects strictly more attacks than either alone, with $A_{\text{static}} \setminus A_{\text{runtime}} \neq \emptyset$ and $A_{\text{runtime}} \setminus A_{\text{static}} \neq \emptyset$.

Witness partition under canonical PG-DSL (post-Option B). Per `dec_01KR49FTB2ND38DW36094G95ZM` and Mission 3B’s empirical T3 partition (`mission_3b/results/t3_partition_canonical.json`):

$$\begin{aligned} A_{\text{static}} \setminus A_{\text{runtime}} &= \{A_2\} \quad (\text{magnitude poisoning — caught by single-tool DT verifier;} \\ &\quad \text{INVARLLM misses sub-noise-floor underdose}) \\ A_{\text{runtime}} \setminus A_{\text{static}} &= \{W_1, W_2\} \quad (\text{stealth drain + post-admission spoofing —} \\ &\quad \text{outside ordered-pair admission scope}) \\ A_{\text{static}} \cap A_{\text{runtime}} &= \{A_1, A_3, Z_1, Z_2, Z_3\} \\ A_{\text{static}} \cup A_{\text{runtime}} &= \{A_1, A_2, A_3, W_1, W_2, Z_1, Z_2, Z_3\} = 8/8 \text{ canonical attack set} \end{aligned}$$

Both set differences are non-empty; the strict-superset claim of T3 is preserved. Z_3 joins the intersection (caught by both canonical admission and INVARLLM) rather than the runtime-only column it would occupy under the Mission 1A–2C single-tool-only admission. The Mission 3A tractable verification (§??, §??) confirms Z_3 sits in the detected region of $T1_v$.

6 Numerical Verification

`mission_3a/t1_v.verify.py` samples composition attacks with parametrized $\|\delta_v\|_\infty$ on the two-tank tractable model (`mission_3a/tractable_model.v.md`); for each magnitude bin it runs the lifter $\rightarrow$ DT $\rightarrow$ vectorized matcher pipeline, recording the empirical detection rate. `mission_3a/results/detection` overlays the empirical curve on the theoretical $1 - F_v(\tau, \varepsilon_{L,v}, \varepsilon_{DT,v})$ step under bounded-error sensing. The empirical curve sits at or above the bound for all τ , confirming non-vacuity. The $Z3$ witness lies far inside the detected region ($\|\delta_v\|_\infty = 4.5 \gg \delta_v^* = 0.432$). `mission_3a/results/t1_v.witnesses.json` records concrete $(\tau, \varepsilon_{L,v}, \varepsilon_{DT,v})$ instances for both the detected and undetectable regimes.

7 Cross-References

- Brain $T1_v$ statement source-of-truth: `jrn_01KR4F6V8RD7YJJ55Z13JBMQXZ`
- Z3b option-B canonical decision: `dec_01KR49FTB2ND38DW36094G95ZM`
- Mission 3B canonical pipeline: `mission_3b/admission_layer/`
- Mission 3B canonical results: `jrn_01KR4BGR32R2XX6PK671WQER7B`
- Mission 1A T1 (consistency check at $n = 1$): `mission_1a/theorems.tex` §T1
- Mission 1A tractable model: `mission_1a/tractable_model.md`
- Two-tank tractable model: `mission_3a/tractable_model.v.md`